

Fostering Self-Regulated Learning in Asynchronous Undergraduate STEM using Interactive H5P Content

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Abstract

Enhancing student motivation and fostering self-regulated learning are ongoing challenges in online education. Extensive research confirms the importance of self-regulated learning (SRL) in online student success. This study investigates the potential of the H5P interactive learning tool to promote self-regulated learning in an online introductory undergraduate chemistry course, reporting the alignment between directly measured self-regulated learning behaviors and learner perspectives on their self-regulated learning. Analysis of student survey data and H5P interaction metrics showed that students reported positive self-regulated learning behaviors, with H5P particularly enhancing cognitive engagement. Direct measures showed strong Forethought behaviors and significant study regularity. These results indicate that H5P can support specific aspects of self-regulated learning in asynchronous courses but also show discrepancies between perceived and observed SRL behaviors. The lack of correlation between direct and indirect measures warrants further investigation and suggests a need for multi-method research approaches. These promising outcomes demonstrate the potential of interactive learning tools like H5P in enriching the online learning experience and nurturing students' academic success.

Keywords: online education, self-regulated learning, educational technology tools, learner perspectives, undergraduate

H5P in an Asynchronous Online Undergraduate STEM Course: self-regulated learning through an educational technology tool

Especially in recent years, there has been an increase in online undergraduate course offerings in STEM, including the asynchronous online modality. Effective online courses cannot simply be a mirror of a traditional course; the teaching strategies are not always transferable and thus new pedagogical strategies are required. There are some arguments against using traditional lectures in STEM courses for a variety of reasons, including the lack of development of scientific literacy skills (Gardner & Belland, 2012). Furthermore, the online learning environment offers unique options beyond static lecture videos or lists of assigned readings (exposition-centered methods). Instead, course designers can opt for strategies and platforms to promote student engagement in the asynchronous learning environment, including active, constructive, and interactive learning.

Aligned with constructivism, when communication of concepts in a STEM course engages students, learning is more effective (Bada & Olusegun, 2015; Cotner et al., 2013; Freeman et al., 2014). Active engagement occurs when students interact with the content but do not generate new material or information (Prince et al., 2020). Constructive engagement occurs when an activity generates material based on the information received (Prince et al., 2020). Interactive engagement occurs when an activity involves an exchange between two or more students, with both contributing (Prince et al., 2020). In the asynchronous classroom, educational technology can be used to promote student engagement (active, constructive, or interactive) during the presentation of information or by reinforcing learning (Gardner & Belland, 2012).

The H5P platform is an educational technology content creator tool that allows for presentation of information in a way that keeps students engaged as they progress through the lesson. H5P offers a variety of activities, including branching scenarios, crosswords, flashcards, and interactive slides and videos. H5P can be used to provide automatic formative feedback through a variety of question types (e.g. fill in the blank, multiple choice, image hotspot, and drag and drop). This platform also allows for “chunking” of course content into subunits, which is important for cognitive load considerations (Belland, 2017). While working through the content presented via H5P, students are actively engaged through human-computer interaction (Shroff et al., 2019). By embedding ungraded formative assessment “knowledge checks” with immediate feedback, constructive engagement is achieved (Prince et al., 2020).

Although it is a relatively new tool, there is some existing information on the use of H5P in undergraduate STEM courses. A hybrid course using H5P for lecture content reported that 60% of students consistently used the H5P course resources (Sinnayah et al., 2021). Compared to traditional lecture content delivery, students reported that the platform required more effort and was more time consuming (Sinnayah et al., 2021). However, most students reported that their knowledge was enhanced by using the H5P resource (Sinnayah et al., 2021).

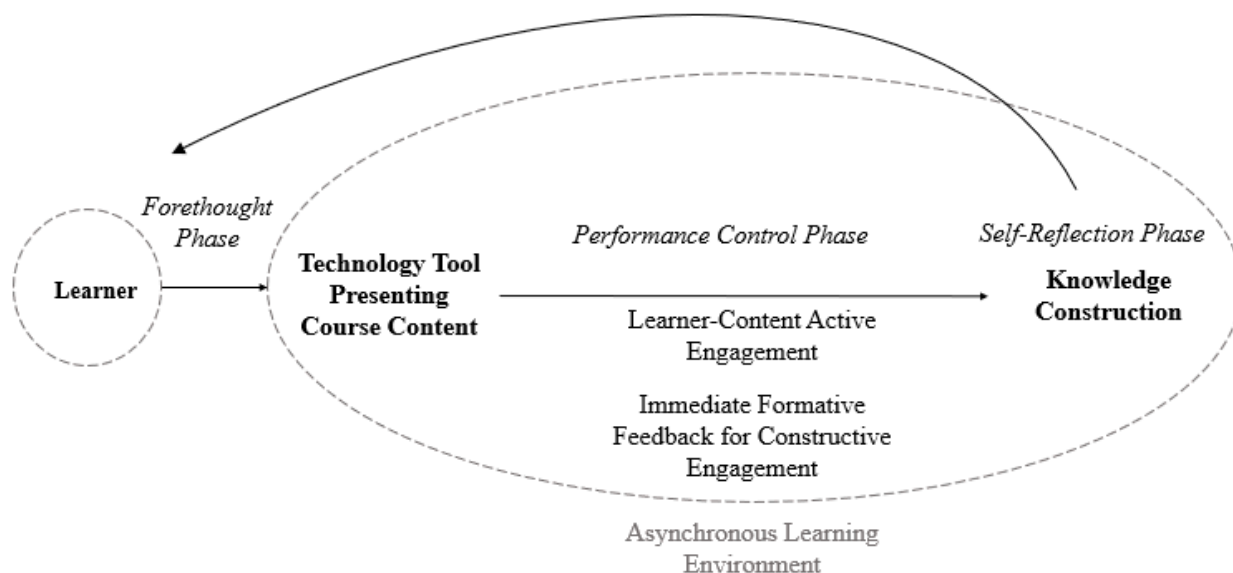
Engaged learning allows students to demonstrate self-regulated learning (SRL) through motivation and participation in their own learning (Corno, 1986; Saks & Leijen, 2014; Zimmerman, 1990). Self-regulated learning can be evaluated according to specific factors, which can vary based on the SRL model, but typically include cognitive, metacognitive, behavioral, motivational, and affective/emotional factors (Alexiou & Paraskeva, 2010; Panadero, 2017). The Zimmerman SRL model is organized into the phases of “forethought”, “performance control”, and “self-reflection” (Zimmerman, 2000). Within undergraduate STEM courses, students that demonstrated certain SRL behaviors performed better on summative assessments (Lawanto et al., 2017). For example, irregularity of interval of access of the learning tool predicts performance (Jo et al., 2014).

To conceptualize how self-regulated learning operates within the specific context of an asynchronous, technology-mediated environment, we apply an adaptation of the Zimmerman model. Figure 1 presents this conceptual framework, illustrating the iterative process students undertake as they interact with course content and receive feedback. In this framework, the performance control phase represents the active engagement with the H5P resource, where students employ learning strategies, monitor their progress, and receive immediate formative feedback through the embedded knowledge checks that serve as a key input as the learner moves to the self-reflection phase. When summative feedback is received on the assessment, the cycle loops back to the next cycle of forethought, demonstrating how the asynchronous H5P tool encourages self-regulated learning.

Figure 1

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Conceptual Framework of Engaged Learning through Information Presentation via Technology Tool in Asynchronous Online Courses



In this study, we want to understand student opinions related to the use of an educational technology tool that promotes active and constructive engagement and self-regulated learning. Importantly, we also want to conduct a direct, empirical comparison to explore the alignment between student perceptions of self-regulated learning (an indirect measure) and observation of self-regulated learning behaviors (a direct measure). This is a gap in the current self-regulated learning literature. While studies often rely on one or the other, it is unclear how well these perceptions align with actual behavior. The research questions are:

1. Do students have a positive opinion of the H5P tool for presenting course content?
2. Do students using H5P resources demonstrate self-regulated learning behavior(s)?
3. Does a learner's perspectives of their self-regulated learning predict measurable self-regulated learning behaviors?

Methods

H5P Design

The H5P interactive resources implemented in the course were aligned with the course learning outcomes of the introductory general chemistry course. Individual H5Ps were created to chunk information aligned with module learning objectives which can reduce cognitive load and improve the context of the information. Content slides were designed with a modified version of Assertion Evidence presentation style to reduce wordiness and emphasize key points (Figure 2).

Figure 2

Example of content slide in H5P

Lewis structures are used as a simple way to show valence

Valence electrons are represented by *dots*

Lone pairs of electrons are represented as *two dots together*

	It is a noble gas so it has a full octet. There's an imaginary box drawn around Ne for placing the electrons.
	You can predict this from its periodic group or you can look at its electron configuration.
	Notice you fill one on each side before you start pairing.

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Learner-content engagement was promoted through use of tools such as image hotspot, advanced fill in the blank, and drag and drop features in H5P (Figure 3). Each interactive component was programmed to provide immediate wrong answer feedback. H5P allows learners to investigate their understanding by using the “check”, “retry”, and “show solution” options.

Figure 3

Example of interactive element in H5P

What is the molar mass of sodium carbonate?

Grams $\longleftrightarrow$ Molar Mass $\longleftrightarrow$ Moles $\longleftrightarrow$ Avogadro's Number $\longleftrightarrow$ Particles

Formula: Na_2CO_3

From the periodic table:
 Na = 23 amu = 23 g/mol
 C = 12 amu = 12 g/mol
 O = 16 amu = 16 g/mol

$(23 \times 2) + (12 \times 1) + (16 \times 3) =$ g/mol

Note: Need a hint? Click the purple speech bubble at the bottom corner of the task.

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The H5P resources were presented to students within the Learning Management System (LMS) on a multimedia resources page for each module, separated into tabs. The first tab is titled “Getting Started” and includes the H5P(s) and instructor-generated calculation video(s). The next tab is titled “Digging Deeper” and lists the relevant sections of the textbook and provides links to related textbook resources like animations. The final tab is titled “Expanding Your Knowledge” and offers a location for the instructor and students to share external resources they found relevant and helpful.

Implementation of H5P

While there are three types of integrations (direct link or embed from H5P.com, learning tools interoperability (LTI) integration, or H5P plugins), the courses in this study used LTI integration with the learning management system Canvas. This approach allows for easy instructor access to learning analytics (Drill Down Report) and the option to embed participation into the gradebook (ungraded). The H5P resources were not assessed nor were they mandatory.

Population & Sample

This study was reviewed by the Institutional Review Board and deemed exempt (Approval #22-115). The target population was General Chemistry I students enrolled between August 2022 and May 2023. The sample (n=13, 13.1% response rate) was drawn from the population through self-selection sampling. Participants were asked to complete a one-time confidential survey. Completion of this survey also provided consent for using performance and learning analytics data obtained from the LMS. Participation was incentivized (\$5 Amazon e-card).

Data Collection

Indirect Measures of Self-Regulated Learning

Participants were asked to respond to a voluntary confidential survey regarding their perspectives on the H5P course resources and their learning. The survey covered basic demographic data and included two research sections. The first section used the Motivation and Self-Regulated Learning Questionnaire (MSLQ), which has been validated in multiple studies, including STEM and undergraduate student populations (Jackson, 2018; Magno, 2011; Pintrich & Groot, n.d.) as well as asynchronous online student populations (Cho & Summers, 2012). The MSLQ measures motivational beliefs (self-efficacy, intrinsic value, and test anxiety) as well as SRL strategies (cognitive strategy use and self-regulation). The MSLQ is a very popular choice for researchers measuring SRL (Panadero, 2017). The second section used the Reduced Instructional Materials Motivation Survey (RIMMS), which has been validated in multiple studies (Loorbach et al., 2015; Becerra & Almendra, 2020; Wang et al., 2020). This model measures four motivation factors: attention, relevance, confidence, and satisfaction.

Direct Measures of Self-Regulated Learning

Direct measures of self-regulated learning were obtained through analysis of learner interactions with H5P resources and other elements within the LMS. H5P analytics, accessed via Drill Down Reports, provided data on access dates, durations, and other relevant metrics. The research team used existing dimensions of SRL in the research literature to identify learner behaviors and actions associated with each category of the SRL framework, which outline cognitive, metacognitive, motivational, and affective components. Forethought SRL can be evaluated by studying initiation (how close to the deadline access to the resource began); performance control SRL can be evaluated by time investment (e.g. time spent with a resource) and study regularity (e.g. resource access frequency); self-reflection SRL can be evaluated by resource access after an attempting the summative assessment and receiving feedback (Kim et al., 2018). The research team developed a comprehensive list of specific learner actions to assess direct measures of SRL connected to the use of H5P within the online chemistry course (Table 1).

Table 1

Direct Measures of Self-Regulated Learning

Category	Subcategory	Learner Actions
Forethought	Initiation	Time between deadline and first H5P access
	Planning & Preparation	Time between first H5P access and first attempt on summative assessment
Performance Control	Time Investment	Total time using H5P resource
	Study Regularity	Resource access frequency
	Persistence & Effort	Whether re-attempted ungraded H5P knowledge check after failing to achieve mastery
Self-Reflection	-	Resource access after first attempt on summative assessment

Data Analysis

Given the small (n=13) sample size for the Likert-scale responses for the MSLQ and RIMMS surveys as well as both surveys being validated as relating specific questions to latent factors, survey responses were grouped and analyzed by factor. This led to 9 factors being analyzed: self-efficacy, intrinsic value, test anxiety, cognitive strategy use, and self-regulation (MSLQ factors) along with

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attention, relevance, confidence, and satisfaction (RIMMS factors). MSLQ factors asked students to rate their agreement with statements *about themselves* on a scale of 1 (not at all true of me) to 7 (very true of me) while the RIMMS factors asked students to rate their agreement with statements *about the H5P resources* on a scale of 1 (not true) to 5 (very true). We analyze the results of the survey in two ways: presenting frequency distributions of responses for MSLQ and RIMMS factors separately and a correlation matrix between MSLQ factors and RIMMS factors.

Kim et al. (2018) have previously measured SRL within a higher education context and undertaken a comprehensive factor analysis, revealing that metrics associated with forethought, performance control and self-reflection loaded onto distinct factors. To determine if this model fits our data, we completed a confirmatory factor analysis using the *lavaan* package for R (Rosseel, 2012), based on the model previously described (Kim et al., 2018). The four-factor model was estimated using the Maximum Likelihood (ML) estimator on a total of 137 observations. The quality of the model fit was assessed based on the χ^2 statistic, degrees of freedom, *p* value, root mean square error of approximation (RMSEA), standardised root mean square residual (SRMR) and the comparative fit index (CFI), using cutoff thresholds reported by Hooper et al. (2008).

Results

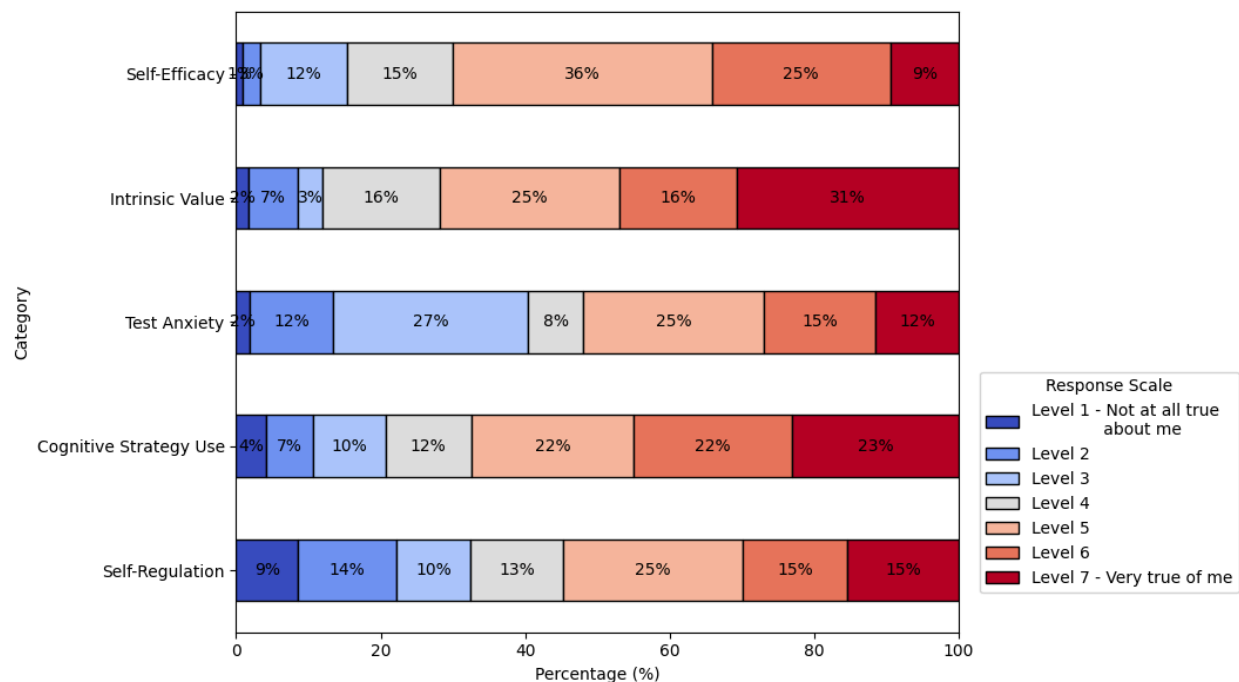
MSLQ Factors (Indirect SRL Measure)

Frequency distribution, presented as percentage to normalize across factors with different numbers of questions, are presented in Figure 4. All 5 factors present a skewed-left distribution centered around 5. As the Likert scale for these questions were between 1 and 7, this suggests students overall strongly agreed with:

- Having high test anxiety;
- The ability to motivate oneself to work on course content;
- Efficacy in study skills and ability to complete content;
- Interest in the course content; and
- The ability to effectively study for the course assessments.

Figure 4

Student Self-Perceptions of Self-Regulated Learning Measured via MSLQ Subscales



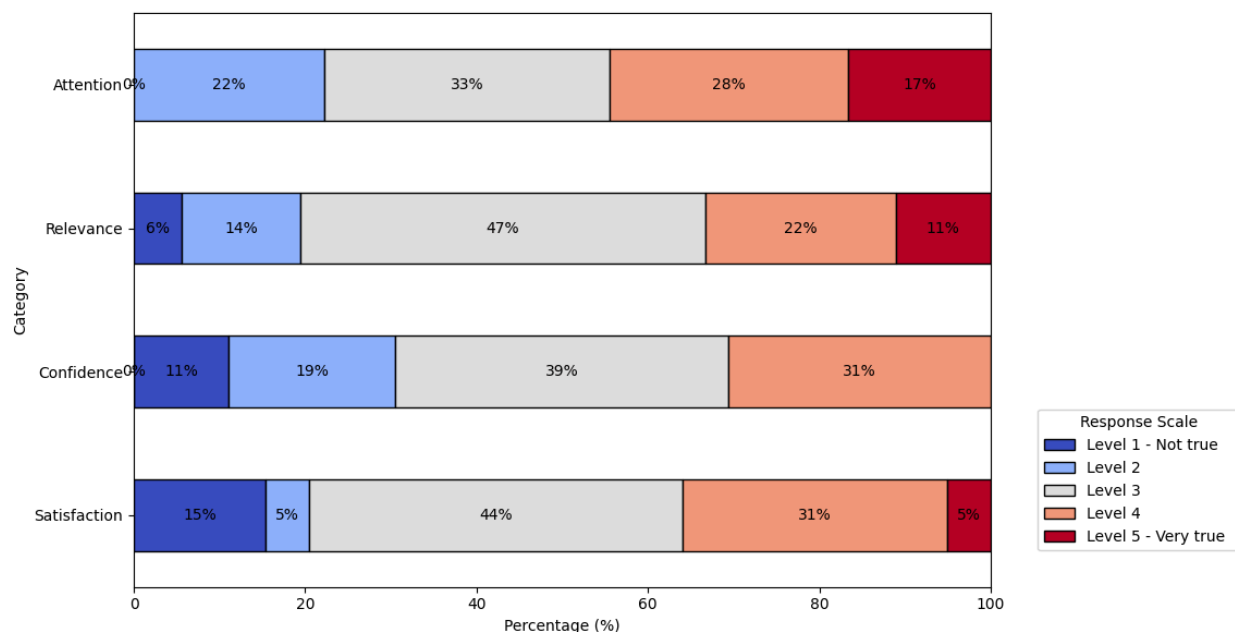
RIMMS Factors

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The RIMMS instrument assessed student perceptions of the H5P resources across the factors of attention, relevance, confidence, and satisfaction, with responses collected on a 5-point Likert scale. Frequency distribution, presented as percentage to normalize across factors with different numbers of questions, are presented in Figure 5. We see three different distributions centered around a response of 3: approximately normal (relevance and satisfaction), skewed-right (confidence), and skewed-left (attention). These results suggest students felt strongest that the H5P decreased their confidence in learning the course content and answering related questions, while the H5P increased their attention to the content. Overall, students' perceptions of the H5P on relevance and satisfaction were mixed.

Figure 5

Student Self-Perceptions of Instructional Materials Motivation Factors Measured via RIMMS Subscales

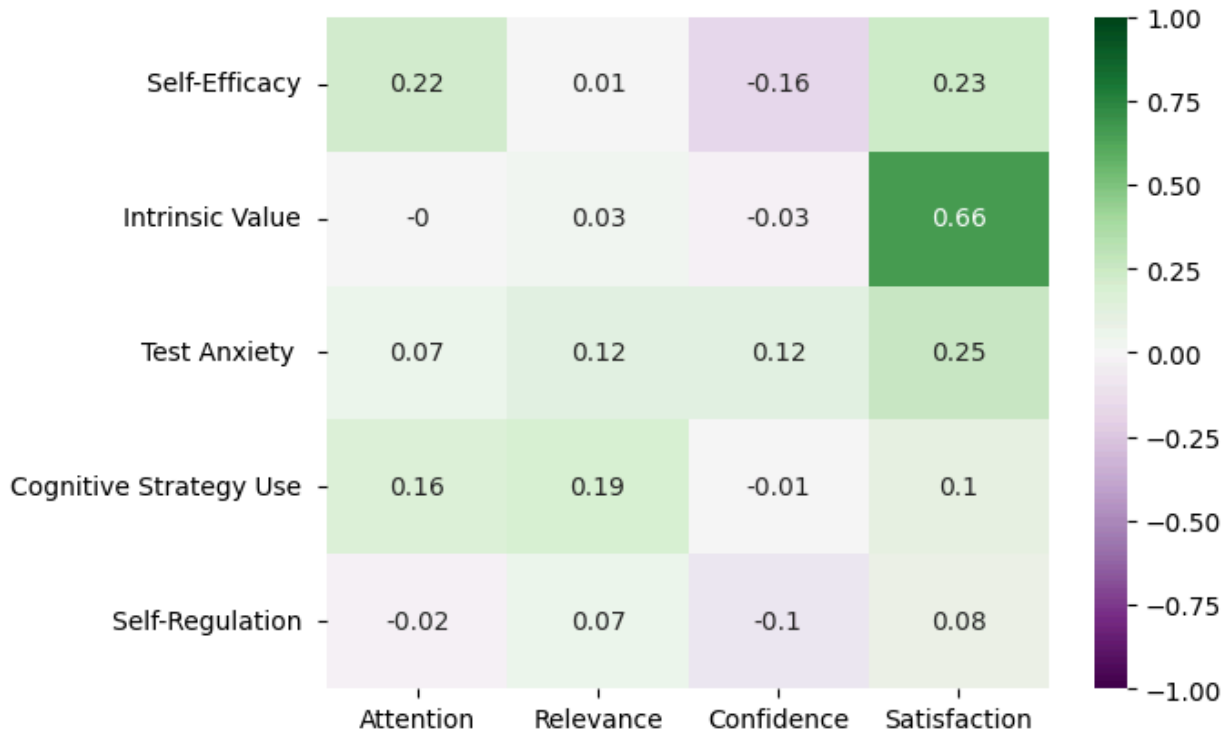


Correlation Between Indirect SRL Measure (MSLQ) and RIMMS Factors

A Pearson correlation coefficient was calculated between each MSLQ and RIMMS factor (Figure 6). Correlation coefficients take values between -1 and 1 where the absolute value of the correlation coefficient represents the strength of the correlation between two factors while the sign of the correlation coefficient represents the direction of the correlation. For example, a correlation coefficient of -0.5 would suggest that as one factor's responses increase by 1, the other factor's responses decrease by 0.5. The MSLQ factor "Intrinsic value" showed moderate positive correlations with three RIMMS factors "Attention" (+0.59, $p < 0.001$), "Relevance" (+0.58, $p < 0.001$) and "Satisfaction" (+0.64, $p < 0.001$). All other associations returned weak or negligible correlations. Students who reported that the course content was interesting and useful to them felt that the H5P activities helped to maintain their interest and were relevant to their learning goals. In addition, students who held high levels of intrinsic value in their studies, reported higher levels of satisfaction when using the H5P activities.

Figure 6

Correlation Coefficient Matrix Relating MSLQ and RIMMS Factors



Behaviour Observation (Direct SRL Measure)

A confirmatory factor analysis was undertaken to confirm the value of the H5P direct measures of SRL used in this study to previously defined SRL factor analysis (Kim et al., 2018, Tang et al., 2024). Based on these prior studies, we propose a four-factor model for our data. We hypothesize that metrics associated with Forethought ('Initiation', 'Planning & preparation') will load onto the same factor (Kim et al. 2018). Metrics associated with Performance control will be split, with the metrics 'Time investment' and 'Persistence & effort' loading onto a single factor (Kim et al., 2018), while 'Study regularity') loading onto a different factor (Tang et al., 2024). Finally, based on these models, it was hypothesized that 'Self reflection' will load onto a unique factor (Kim et al., 2018, Tang et al., 2024).

The proposed four-factor model demonstrated a good fit to the data. Key goodness-of-fit indices (Table 2), exceed commonly accepted thresholds for a good model fit (Hooper et al., 2008). The standardised covariances between latent variables (correlations) Factor 1 (Forethought) and Factor 2 (Self-reflection) showed a significant ($p < 0.01$) negative correlation (-0.31), suggesting a weak, inverse relationship (Figure 4). Furthermore, a significant ($p < 0.01$) positive correlation (+0.32) between Factors 2 (Self-Reflection) and 3 (Performance control) was observed (Figure 7). The observed variables, 'Initiation', 'Planning & Preparation', 'Self-reflection' and 'Study regularity', showed significant correlations ($p < 0.5$) with their respective latent variables (Figure 4). In contrast, observed variables 'Time Investment' and 'Persistence & Effort' showed insignificant correlations ($p = 0.839$ and $p = 0.834$, respectively) on Factor 4, which indicates that 'Time Investment' and 'Persistence & Effort' are not good predictors of SRL in our study. While these factors represent poor measures of SRL behaviour, we have deliberately included them here for completeness and consistency with the hypothesized SRL framework.

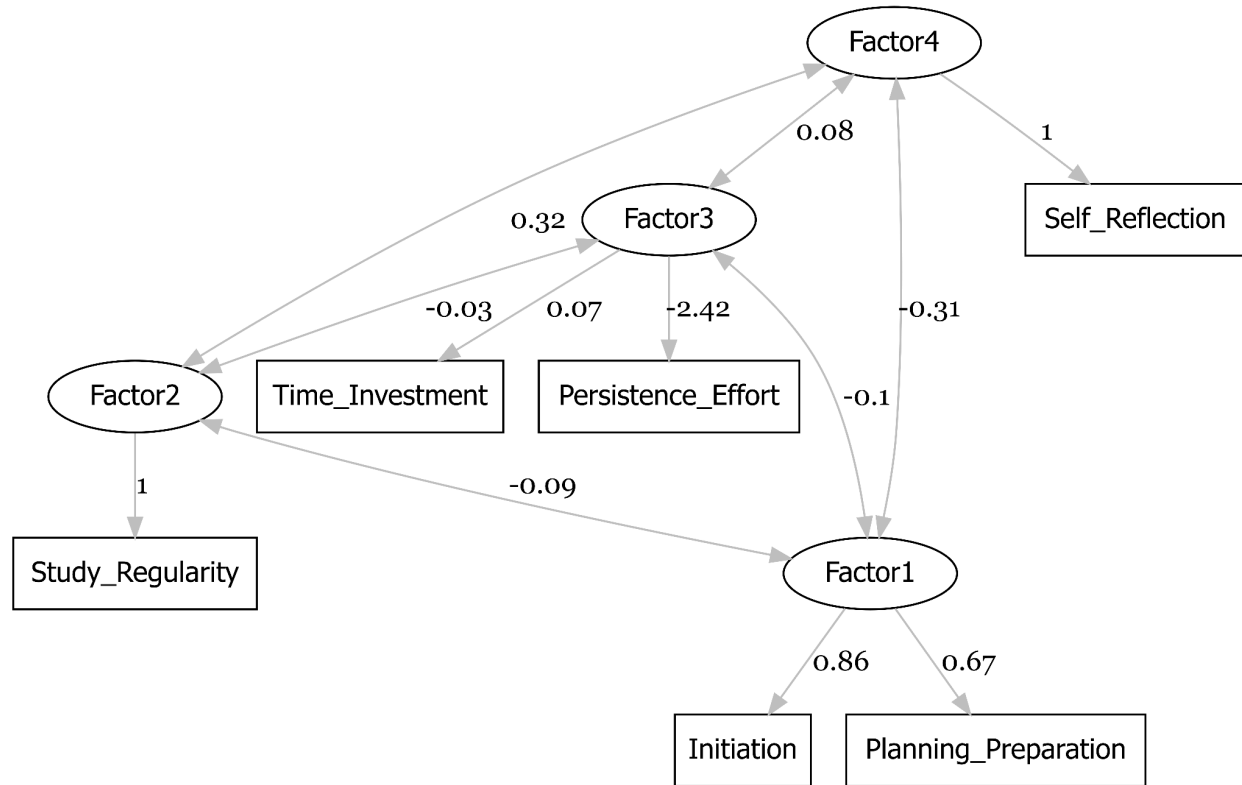
Table 2

Goodness-of-fit indices demonstrate a good fit of our proposed model to the data.

Model	X ²	df	p-value	CFI	RMSEA (95% CI)	SRMR
Hypothesized model	6.99	5.00	0.222	0.978	0.054 (0.00, 0.139)	0.033

Figure 7

CFA Path Diagram for the SRL Framework Illustrating Relationships Between Latent SRL Phases and Observed Behavioral Measures. Latent factors are represented as ovals and represent the three main phases of the SRL framework (Zimmerman, 2000) - Forethought, Performance control and Self-regulation. Rectangles represent the direct measures (observed variables) of SRL. Arrows that connect latent factors to the direct measures are labelled with the correlation coefficient, all of which were significant except for Time Investment and Persistence and Effort. Double-headed arrows connecting latent factors represent the colinearity between factors, depicting the interrelatedness between factors.



Correlation Between Direct SRL Measure (behavior observation) and RIMMS Factors

In this comparison of the direct measure of self-regulated learning through observation of behaviours to the self-reported measures of students attitudes and perspectives regarding the H5P tool. There were only negligible correlations, none significant. Variance in the direct measure data compared to the indirect measure data limited the analysis and the sample size was too small for other statistical measures.

Direct Versus Indirect Measures of Self-Regulated Learning

In the comparison of direct versus indirect measures of self-regulated learning, the correlations were negligible, with none significant. A larger sample size could overcome the variance observed in the metrics used

Discussion

RQ1 – Learner Opinion of H5P Tool

Learners' opinions of the H5P tool were mixed or neutral. Based on the frequencies presented on survey items associated with RIMMS factors, students were split on Satisfaction (20% negative, 43% neutral, 35% positive) and Relevance (18% negative, 47% neutral, 33% positive), skewed negative for Confidence (30% negative, 38% neutral, 30% positive), and skewed positive for Attention (22% negative, 33% neutral, 43% positive). The largest category for both Satisfaction and Relevance was the neutral

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response (43% and 47%, respectively), suggesting widespread indifference or unwillingness to clearly endorse a directional response on these factors, either positive or negative. This is offset partially by the positive skew in Attention, with 43% positive response, suggesting that the interactive elements in H5P were effective at capturing and holding student interest.

It is important to note the +0.66 correlation between Intrinsic Value (MSLQ) and Satisfaction (RIMMS), which suggests the most satisfied students were the ones who viewed the course content as interesting and valuable. This suggests that the H5P tool enhanced the experience for already motivated learners but did not shift the opinion of unmotivated or indifferent learners. Overall, these results on learner opinion on the H5P tool align with previous research where students reported increased learning effort, but also an increase in content knowledge mastery (Sinnayah et al., 2021).

RQ2 – Demonstration of SRL using H5P

Learners' self-perceptions of their own self-regulated learning capacity while using H5P were largely positive across all MSLQ factors, ranging from the highest for Intrinsic Value (70%) to the lowest for Test Anxiety (51%). However, this study sought to also directly measure self-regulated learning. Confirmatory Factor Analysis validated the structural integrity of specific behavioral measures derived from H5P use, confirming that the specific metrics aligned with the three-phase SRL framework (Zimmerman, 2000). This provides empirical insight into what actions leave digital trace data that capture SRL activity.

Forethought reflects the first phase of the SRL framework, characterised by breaking down the task and setting specific goals for their performance (Zimmerman, 2000). We mapped the H5P metrics "Time between deadline and first H5P access" (Initiation) and "Time between first H5P access and first attempt on summative assessment" (Planning & preparation) as variables to measure Forethought in learner behaviours. We found that these output variables demonstrated strong, significant ($p < 0.01$) factor loadings on Factor 1 (Forethought), indicating that these variables represent a good predictor of SRL initiation in this asynchronous learning environment. These findings align with previous studies where the importance of goal setting and breaking tasks down into their constituent parts has been associated with higher SRL rating and improved academic performance (Lawanto, 2017, Kim et al., 2018, Sun et al., 2023).

Performance control, measured by the H5P metric "Study regularity", proved to be a significant predictor of SRL activity, but the proposed Performance control metric "Total time using H5P resource" was a weak predictor of SRL activity. This emphasizes that consistent, routine access is a more notable indicator of SRL than simple time investment. "Whether re-attempted ungraded H5P knowledge check after failing to achieve mastery" was found to be an insignificant predictor of SRL activity. These findings align with previous research, where the amount of time invested in a task and the number of task re-attempts are not necessarily reflected in the learner's SRL activity nor their academic performance (Kim et al., 2017; Sun et al., 2023; Kong & Lin, 2023).

Self-reflection, measured in our study using "Resource access after first attempt on summative assessment", showed a weak positive correlation with Performance control ($p < 0.01$). This suggests that the H5P activities function as a form of "fact-checking" or knowledge gap analysis supported by the tool's availability. Interestingly, a weak negative correlation with Forethought factors ($p < 0.01$) indicates that learners' Forethought strategies were inversely related to personal reflection. This finding is in contrast with Zimmerman's (2000) cyclical framework that argues reflection should inform future planning. This observed inverse relationship may reflect a cohort-specific behaviour. This theoretical conflict warrants further investigation.

RQ3 – Alignment between Direct and Indirect Measures of SRL

The MSLQ served as the indirect measure of self-regulated learning, capturing metacognitive beliefs and motivational orientations. The direct observations derived from students' interactions with the H5P tool and LMS served as objective indicators of actual enacted behaviors during the performance control and self-reflection phases. We propose that students who report high use of strategies for effective regulation in the MSLQ should demonstrate those strategic behaviors in their H5P use metrics. In this comparison, we can assess convergent validity between the learner's self-awareness and their observable self-regulated learning actions. Based on our study, we found no significant correlations between the indirect and direct measures of SRL. As such, we are unable to infer motivational or satisfaction levels based on direct H5P use metrics.

If a correlation is found, researchers can have greater confidence in the predictive validity of the indirect measure approach, which is easier to administer. However, if divergence is observed, it draws self-reported data for SRL into question as insufficient or biased, compelling future researchers to use mixed methods or design for triangulation.

Implications for Practice

This study validates that, while H5P supports self-regulated learning, practitioners must prioritise the quality of engagement over simple metrics of effort. Initiation and Study Regularity were strong, significant predictors of Forethought and Performance Control, respectively. This validates the use of H5P trace data to reward or coach students towards early access and consistent engagement. However, Time Investment and Re-attempts proved to be insignificant predictors of SRL activity. This suggests that simply providing students unlimited time or attempts is not enough to promote effective self-regulation. Instructional design choices must be made to scaffold *how* students use multiple attempts, rather than simply offering them.

The concept of providing multiple attempts coupled with immediate feedback is a core feature of H5P that aligns with active learning principles that promote engagement, participation, and metacognition (e.g., Faulconer et al., 2021). However, the data presented in this study shows a weak, inverse correlation between Forethought and Self-Reflection. This may suggest either that students who plan poorly are forced into self-reflection or it may reflect a limitation in the structural support for a true reflective loop within this specific asynchronous learning environment. Online educators could design instructions and scaffolding to force a deliberate reflective loop and drive learners towards help-seeking behaviours before proceeding (Yang & Stefaniak, 2023; Onder & Akcapinar, 2023; Brazhkin & Strakos, 2023). Sequential SRL training could explicitly scaffold the self-reflection to forethought link.

Future qualitative research could move beyond the inverse correlation between Forethought and Self-Reflection to explore the 'why' through post-hoc interviews or SRL journals. This inquiry could isolate if this observed inverse relationship is a design failure or a consistent behavioral characteristic of online learners in this context.

Using these insights, educators can create more effective and student-centered active learning opportunities for online learning, ensuring that the H5P tool is a powerful instrument for developing deliberate and effective self-regulation, contributing to enhanced educational outcomes.

Limitations

This study had several limitations that may impact the interpretation and generalizability of these findings. These include sample size, self-selection sampling, and scope of the case study. These warrant caution in drawing definitive conclusions and point to areas where additional investigation is appropriate.

The study had a limited sample size, limiting generalisability. Nonresponse error might be influenced by sample size. In this study, we weighed the reduction in nonresponse error from increasing contact attempts with the potential increase in measurement error (Kreuter et al., 2010). Self-selection bias can impact the representativeness of the sample in education studies, though the impact can vary based on context and methodology (Hernandez, 2015).

While validated instruments were used in this study, there may be limitations of these tools in capturing the complex nature of self-regulated learning and student perceptions.

This case study focused on a single course within a discrete time frame, limiting the generalizability of the findings to other STEM and non-STEM disciplines or academic levels. As this case study was also a snapshot investigation of perceptions, it inherently did not capture longitudinal insights into self-regulated learning behaviors and perceptions.

Conclusion

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This study investigated the potential of H5P as an educational technology tool to enhance self-regulated learning in an asynchronous online undergraduate STEM course, using both direct and indirect measures. Key findings from this study are:

- While learner opinions of the H5P tool were mixed, a clear strength of the tool highlighted in this data is the ability to improve their attention to the material. Satisfaction with the H5P tool was not universal, but it was linked to prior motivation, with the highest satisfaction reported by students who perceived the course content as having high intrinsic value.
- Students generally self-reported positive self-regulated learning behaviours associated with H5P use (MSLQ). Self-regulated learning behaviors were observed in digital trace data. The negative correlation between Forethought and Self-Reflection highlights a conflict with existing theory, challenging an inherent cycling model for self-regulated learning in this particular asynchronous learning environment's design.
- There weak alignment between direct and indirect measures of self-regulated learning when using H5P warrants further investigation with an increased sample size. This also underscores the importance of employing a multi-method approach to gain a holistic understanding of SRL in online learning environments.

This study establishes that while H5P is a viable digital tool for supporting self-regulated learning, its effectiveness is contingent on design and scaffolding to ensure the cyclical SRL loop is achieved. By identifying key SRL behaviours and comparing direct versus indirect SRL measures, this study provides a new framework for SRL measurement in asynchronous learning contexts. Future research should focus on the efficacy of structural interventions in this space, aiming to close the observed inverse loop between forethought and self-reflection. This will ensure that digital tools transition from merely enabling some SRL behaviours to actively supporting the entire self-regulatory cycle.

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Declaration of Conflicting Interests

Three authors wish to declare current/former employment at the studied institution as a potential conflict of interest.

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